

B.E. II EXAMINATION NOV-DEC' 2022

Information Technology

3AIRC1: Discrete Mathematics

Duration: 3 Hrs]

[Max. Marks: 60

Note: Attempt any TWO parts from each question. Each carries equal marks.

- Q1. (a) Determine the number of integers between 1 to 150 that are divisible by any of the integers 2, 5, 7 and 11. (06)
- (b) (i) For the following hashing function, show how the following data would be inserted in the order given in initially empty cells. Use the usual collision resolution policy to resolve collision, if occurs: $h(x) = x \pmod{11}$, cells indexed from 0 to 10; the data is: 55, 13, 281, 743, 377, 20, 10, 796 (03)
- (ii) Construct the Hasse diagram (writing all steps properly) for the partial ordering $\{(a, b): a \text{ divides } b\}$ on the set $\{2, 3, 5, 9, 12, 15, 18\}$. (03)
- (c) Use Warshall's Algorithm to find the Transitive Closure of the following relation R on the set $A = \{1, 2, 3, 4\}$: (06)
- $$R = \{(1, 1), (1, 4), (2, 1), (2, 3), (3, 1), (3, 2), (3, 4), (4, 2)\}$$
- Q2. (a) Let p, q, r be the propositions: p : Grizzly bears have seen in the area, (06)
- q : Hiking is safe on the trail, r : Berries are ripe along the trail.
- Write these propositions using p, q and r and logical connectives:
- (1) Berries are ripe along the trail when Grizzly bears have seen in the area or Berries are ripe along the trail when hiking is safe on the trail.
- (2) Berries are ripe along the trail whenever Grizzly bears have seen in the area and hiking is safe on the trail.
- Hence prove that the above statements are logically equivalent using truth table.
- (b) State the converse, contrapositive and inverse of the following implication: (06)
- "To get the tenure as a Professor, it is sufficient to be world famous".
- (c) (i) Let $F(x, y)$ be the statement "x can fool y" where the domain consists of all people in the world. Use quantifiers to express each of these statements: (04)
- (1) Evelyn can fool everybody.
- (2) Everybody can fool somebody.
- (3) No one can fool both Fred and Jerry.
- (4) Everyone can be fooled by somebody.
- (ii) Express each of these mathematical statements using predicates, quantifiers, logical connectives, and mathematical operators: (02)
- (1) A negative real number does not have a square root that is a real number.
- (2) The absolute value of the sum of two integers does not exceed the sum of the absolute values of these integers.
- Q3. (a) Use mathematical induction to prove that $n^5 - n$ is divisible by 5, whenever n is a non-negative integer. (06)
- (b) Use generating function to solve the recurrence relation $a_n = 5a_{n-1} - 6a_{n-2}$ with the initial conditions $a_0 = 6, a_1 = 30$. (06)